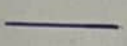
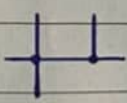
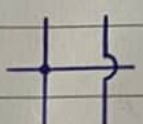
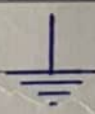
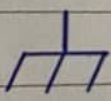


Experiment no. 1

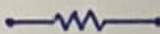
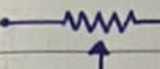
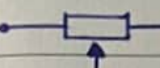
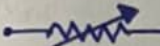
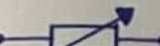
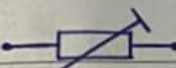
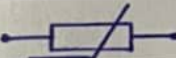
Familiarization of Symbols

Aim:

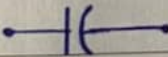
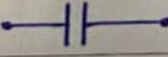
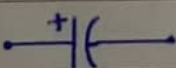
To familiarise and understand the general symbols in electrical and electronic circuit.

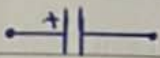
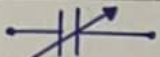
Symbol	Component Name	Meaning
Wire Symbols		
	Electrical wire	Conductor of electrical current.
	Connected wire	Connected Crossing
	Not connected wire	Wires are not connected
Ground Symbols		
	earth ground	used for zero potential (reference) and electrical shock potential
	chassis ground	connected to chassis of the circuit
	digital / common ground	

Resistor Symbols

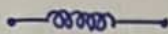
	Resistor (IEEE)	Resistor reduce the Current Flow
	Resistor (IEC)	
	Potentiometer (IEEE)	Adjustable resistor has three terminals
	Potentiometer (IEC)	
	Variable resistor / Rheostat (IEEE)	Adjustable resistor has 2 terminals
	Variable resistor / Rheostat (IEC)	
	Trimmer resistor	Preset resistor
	Thermistor	Thermal resistor - changes resistance when temperature changes
	Photoresistor / light dependent resistor (LDR)	

Capacitor Symbols

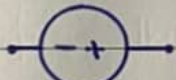
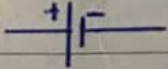
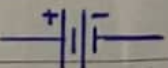
	Capacitor	Capacitor is used to store electric charge. It act as short circuit with AC and open circuit with DC
	Capacitor	
	Polarised capacitor	electrolytic capacitor

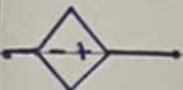
	Polarised Capacitor	electrolytic capacitor
	Variable Capacitor	Adjustable capacitor

Inductor / coil Symbols

	Inductor	coil/solenoid that generates magnetic field
	Iron Core inductor	Includes Iron
	Variable Inductor	

Power Supply Symbols

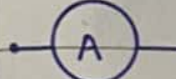
	Voltage Source	Generates constant Voltages
	Current Source	Generate constant current
	AC Voltage source	AC Voltage source
	Generator	electrical voltage is generated by mechanical rotation of the generator
	Battery cell	Generates constant voltage
	Battery	Generates constant Voltage

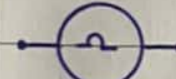
	Controlled Voltage Source	Generates Voltage as a function of Voltage or current of other circuit elements
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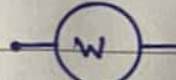
	Controlled Current Source	Generates current as a function of Voltage or current of the other circuit elements
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Meter Symbols

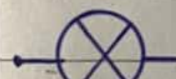
	Volt meter	measures Voltage has high resistance connected in parallel
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	Ammeter	measures electric current has zero resistance connected serially
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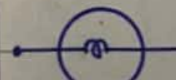
	Ohmmeter	measure resistance
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	Watt meter	Measures electric power
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Lamp/Light Bulb Symbols

	light / lamp bulb	
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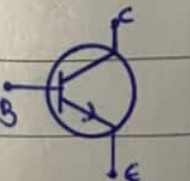
	light / lamp bulb	Generates light when current flows through
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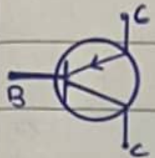
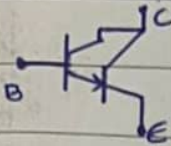
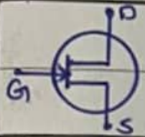
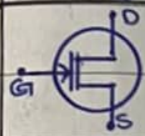
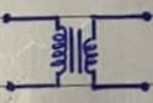
	light / lamp bulb	
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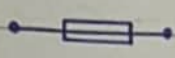
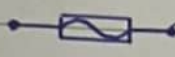
DIODE / LED Symbols

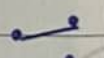
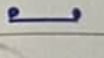
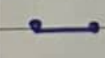
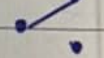
	Diode	Diode allows current flow in one direction only left to right
	Zener diode	allows current flow in one direction but can flow in reverse direction when above breakdown voltage
	Schottky diode	Schottky diode is a diode with low voltage drop
	Tunnel diode	
	Varactor / varicap diode	Variable capacitor diode
	Light Emitting Diode (LED)	LED emits light when current flows
	Photo diode	Photo diode allows current flow when exposed to light

Transistor Symbols

	NPN bipolar transistor	allows current flow when high potential of base (middle)
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	PNP bipolar transistor	Allows current flow when low potential at base (middle)
	Darlington transistor	made of 2 bipolar transistor. Has total gain of the product of each gain.
	JFET - N Transistor	N channel Field effect transistor
	JFET - P Transistor	P channel field effect transistor
	NMOS Transistor	N channel MOSFET Transistor
	PMOS Transistor	P channel MOSFET Transistor
MISC Symbols		
	Motor	electric motor
	transformer	change AC voltage from high to low or low to high
	electric bell	Rings when activated
	Buzzer	produces buzzing sound
Required		

	Fuse	The Fuse disconnect when current above threshold. used to protect circuit from high currents.
	Fuse	

Supplementary Contact Symbols						Terms
SPST NO		SPST NC		SPDT		SPST - Single pole single throw SPDT - Single pole double throw
Single Break	Double Break	Single Break	Double Break	Single Break	Double Break	
						DPST - Double Pole Single throw DPDT - Double Pole Double throw
DSPT NO		DSPT NC		DPDT		NO - Normally Open NC - Normally closed
Single Break	Double Break	Single Break	Double Break	Single Break	Double Break	
						

Result

Familiarised the general symbols in electric and electronic circuits.